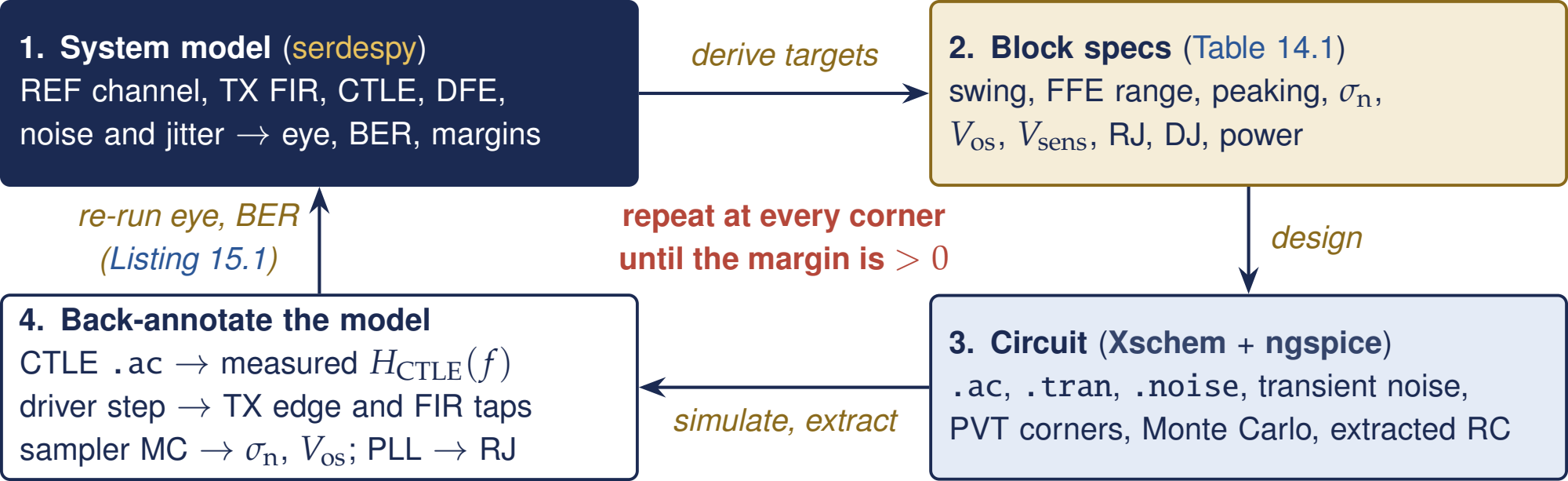

1. System model (serdespy)
REF channel, TX FIR, CTLE, DFE,
noise and jitter → eye, BER, margins
derive targets
2. Block specs (Table 14.1)
swing, FFE range, peaking, σ_n,
V_{os}, V_{sens}, RJ, DJ, power
design
3. Circuit (Xschem + ngspice)
.ac, .tran, .noise, transient noise,
PVT corners, Monte Carlo, extracted RC
simulate, extract
4. Back-annotate the model
CTLE .ac → measured $H_{CTLE}(f)$
driver step → TX edge and FIR taps
sampler MC → σ_n, V_{os}; PLL → RJ
re-run eye, BER
(Listing 15.1)
repeat at every corner
until the margin is > 0